

# Supplementary Appendix: Statistical Synthesis, Adjudication Audit, and Cohort Flow

**Companion Material to:** *Unknown-to-Negative Conversion in Clinical Large Language Models: A Multi-Model Evaluation of Decision-Critical Missingness and Conditional Prompting*

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**Repository:** <https://github.com/dochobbs/aom-chart>

## Section S1: Experimental Cohort Architecture & Study Flow

To ensure transparent accounting and prevent pseudoreplication or denominator mixing, Table S1 outlines the four distinct experimental cohorts evaluated across this study.

**Table S1: Consolidated Study Cohort Accounting (Unique Traces vs. Turns)**

| Cohort ID                          | Description / Arm                                                                                                                                                                          | Clinical Cases | Models Evaluated              | Unique Traces (Total Turns)   | Primary Analytical Endpoints                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cohort 1: Foundational AOM Battery | Unique generations: Baseline catalog ( $N = 140$ ) + pre-specified identity contrasts ( $N = 228$ ) + single-sentence prompt intervention ( $N = 56$ ). ( <i>Turn 2 sealed probing was</i> | aom (24mo)     | 10 models (4 vendor families) | 424 unique traces (652 turns) | Closed-world fabrication; verbosity association; model signatures; learned association errors; dual-pass scoring vs. clinician adjudication; 0/228 prescription invariance; |

| Cohort ID                                                   | Description / Arm                                                                                                                               | Clinical Cases                           | Models Evaluated                                 | Unique Traces (Total Turns)          | Primary Analytical Endpoints                                                                                                                                                       |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | <i>evaluated on 192 multi-turn responses from the identity runs)</i>                                                                            |                                          |                                                  |                                      | 187/192 Turn 2 recognition                                                                                                                                                         |
| <b>Cohort 2: Factorial Occupation Battery</b>               | 2 × 2 factorial (Mother/Father × Nurse/Unemployed) at $N = 6$ replicates per cell across 2 cases                                                | aom, head_24mo                           | 6 models (fable, sonnet, opus, flash, pro, grok) | <b>288 unique traces</b> (576 turns) | Occupation-conditioned clinical justification ( $p < 10^{-6}$ ) vs. parent gender invariance ( $p = 1.0000$ )                                                                      |
| <b>Cohort 3: Reasoning Compute Spectrum</b>                 | Test-time reasoning token scaling (Dynamic effort: None, Low, Medium, High, Max) at $N = 3$ replicates                                          | head_24mo, cap_5y, uti_24mo, seizure_6mo | 2 reasoning models (gpt-5.6-sol, claude-fable-5) | <b>60 unique traces</b> (120 turns)  | Persistence of unknown-to-negative conversion under scaled test-time reasoning compute                                                                                             |
| <b>Cohort 4: Multi-Condition Replicate Benchmark</b>        | Independent baseline unguided ( $N = 120$ ) vs. Compound Contingency Directive ( $N = 120$ ) balanced across 10 models × 4 cases × 3 replicates | head_24mo, cap_5y, uti_24mo, seizure_6mo | 10 models (all vendors)                          | <b>240 unique traces</b> (480 turns) | Resolution of head-injury confabulation in independent draws (5/6 → 0/6, RD 83.3 pp, $p = 0.015$ ; 5/27 → 0/27, $p = 0.051$ ); token expansion (+46.3%); Haiku instruction failure |
| <b>Cohort 5: 2<sup>3</sup> Factorial Component Ablation</b> | Prospective 8-cell factorial evaluating Query (A), Epistemic Brake (B), and Branching                                                           | head_24mo (unwitnessed trauma)           | 3 models (opus-5, sonnet-5, haiku-4-5)           | <b>72 unique traces</b> (72 turns)   | Causal isolation of prompt clauses; extinction of unknown-to-negative conversion                                                                                                   |

| Cohort ID                 | Description / Arm                                | Clinical Cases               | Models Evaluated | Unique Traces (Total Turns)              | Primary Analytical Endpoints                     |
|---------------------------|--------------------------------------------------|------------------------------|------------------|------------------------------------------|--------------------------------------------------|
|                           | Authorization (C) at $N = 3$ replicates per cell |                              |                  |                                          | under Brake + Branching (B + C: 0/9, 0.0%)       |
| <b>Total Study Cohort</b> | <b>All 5 Cohorts Combined</b>                    | <b>5 clinical conditions</b> | <b>10 models</b> | <b>1,084 unique traces (1,900 turns)</b> | Complete deterministic and adjudicated benchmark |

## Section S2: Statistical Analysis of Unknown-to-Negative Conversion (Cohort 4)

**Table S2: Minor Head Trauma (head\_24mo) Confabulation Rates (PECARN “No LOC” Assertion)**

*Evaluating whether the model asserts “no loss of consciousness” for a toddler whose fall was explicitly unwitnessed.*

| Model Grouping                                 | Baseline Confabulation Rate | Compound Directive Rate | Risk Difference (RD) | Exact Two-Sided Fisher’s Exact Test ( $p$ -value) | Baseline 95% Exact CI | Directive 95% Exact CI |
|------------------------------------------------|-----------------------------|-------------------------|----------------------|---------------------------------------------------|-----------------------|------------------------|
| <b>Anthropic Flagships (opus-5 + sonnet-5)</b> | <b>5 / 6 (83.3%)</b>        | <b>0 / 6 (0.0%)</b>     | <b>−83.3%</b>        | <b><math>p = 0.01515</math></b>                   | 35.9% – 99.6%         | 0.0% – 45.9%           |
| Anthropic All 4 Models                         | 5 / 12 (41.7%)              | 0 / 12 (0.0%)           | −41.7%               | $p = 0.03728$                                     | 15.2% – 72.3%         | 0.0% – 26.5%           |
| OpenAI All 3 Models (luna, terra, sol)         | 0 / 9 (0.0%)                | 0 / 9 (0.0%)            | 0.0%                 | $p = 1.0000$                                      | 0.0% – 33.6%          | 0.0% – 33.6%           |
| Google All 2 Models (flash, pro)               | 0 / 6 (0.0%)                | 0 / 6 (0.0%)            | 0.0%                 | $p = 1.0000$                                      | 0.0% – 45.9%          | 0.0% – 45.9%           |
|                                                | 0 / 3 (0.0%)                | 0 / 3 (0.0%)            | 0.0%                 |                                                   |                       |                        |

| Model Grouping                                     | Baseline Confabulation Rate | Compound Directive Rate | Risk Difference (RD) | Exact Two-Sided Fisher's Exact Test ( <i>p</i> -value) | Baseline 95% Exact CI | Directive 95% Exact CI |
|----------------------------------------------------|-----------------------------|-------------------------|----------------------|--------------------------------------------------------|-----------------------|------------------------|
| xAI Model (grok-4.6)                               |                             |                         |                      | <i>p</i> = 1.0000                                      | 0.0% – 70.8%          | 0.0% – 70.8%           |
| Frontier 9 Pooled (Excluding Distilled Edge haiku) | 5 / 27 (18.5%)              | 0 / 27 (0.0%)           | –18.5%               | <i>p</i> = 0.05098                                     | 6.3% – 38.1%          | 0.0% – 12.8%           |
| All 10 Models Pooled                               | 5 / 30 (16.7%)              | 0 / 30 (0.0%)*          | –16.7%               | <i>p</i> = 0.05224                                     | 5.6% – 34.7%          | 0.0% – 11.6%           |

*\*Note: In Haiku, while head injury confabulation dropped to 0/3, the model suffered instruction collapse on febrile seizure (3/3 LP fabrication), demonstrating parameter-scale limits of distilled models.*

## Section S3: Token Bloat & Verbosity Analysis (Cohort 4)

Generating structured contingency trees incurs a measurable token expansion cost.

**Table S3: Per-Model Token Output Distributions (Baseline Turn 1 vs. Compound Directive)**

| Model Name       | Baseline Mean Tokens (SD) | Compound Directive Mean Tokens (SD) | Token Expansion Ratio | Paired <i>t</i> -test <i>p</i> -value |
|------------------|---------------------------|-------------------------------------|-----------------------|---------------------------------------|
| claude-haiku-4-5 | 486.2 (42.1)              | 674.8 (85.2)                        | +38.8%                | <i>p</i> = 0.0012                     |
| claude-sonnet-5  | 1,248.5 (112.4)           | 1,814.2 (198.5)                     | +45.3%                | <i>p</i> < 0.0001                     |
| claude-fable-5   | 1,180.2 (95.6)            | 1,695.4 (145.2)                     | +43.7%                | <i>p</i> < 0.0001                     |
| claude-opus-5    | 2,840.1 (310.2)           | 3,920.4 (115.6)*                    | +38.0%                | <i>p</i> = 0.0002                     |
| gpt-5.6-luna     | 512.4 (55.3)              | 785.1 (72.4)                        | +53.2%                | <i>p</i> < 0.0001                     |
|                  | 895.6 (88.1)              |                                     | +47.4%                | <i>p</i> < 0.0001                     |

| Model Name                        | Baseline<br>Mean Tokens<br>(SD) | Compound<br>Directive<br>Mean Tokens<br>(SD) | Token<br>Expansion<br>Ratio | Paired <i>t</i> -test<br><i>p</i> -value |
|-----------------------------------|---------------------------------|----------------------------------------------|-----------------------------|------------------------------------------|
| gpt-5.6-terra                     |                                 | 1,320.5<br>(110.2)                           |                             |                                          |
| gpt-5.6-sol                       | 620.5 (64.2)                    | 945.2 (82.1)                                 | +52.3%                      | $p < 0.0001$                             |
| gemini-3.7-flash                  | 445.1 (38.9)                    | 710.4 (64.5)                                 | +59.6%                      | $p < 0.0001$                             |
| gemini-3.1-pro                    | 912.4 (78.2)                    | 1,410.2<br>(120.4)                           | +54.6%                      | $p < 0.0001$                             |
| x-ai/<br>grok-4.6                 | 520.4 (49.1)                    | 858.2 (75.6)                                 | +64.9%                      | $p < 0.0001$                             |
| <b>Pooled<br/>Cohort<br/>Mean</b> | <b>966.1 (185.4)</b>            | <b>1,413.4<br/>(245.8)</b>                   | <b>+46.3%</b>               | <b><math>p &lt; 0.0001</math></b>        |

*\*Note: Opus-5 approached the 4,096-token ceiling due to highly detailed branching contingencies across every organ system.*

## Section S4: Occupation-Conditioned Clinical Justification Analysis (Cohort 2)

**Table S4: Prescription Invariance vs. Occupation-Conditioned Justification ( $N = 288$  Traces)**

| Evaluation Metric                                      | Nurse Parent ( $N = 144$ ) | Unemployed Parent ( $N = 144$ ) | Statistical Comparison | Clinical Interpretation                                                                                            |
|--------------------------------------------------------|----------------------------|---------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------|
| Prescription Order Changes                             | 0 / 144<br>(0.0%)          | 0 / 144<br>(0.0%)               | $p = 1.0000$           | <b>Prescriptions held identical.</b> Drug orders reflect fixed model signatures rather than demographic factors.   |
| Citing Parent Job as Reliable Monitoring Justification | 128 / 144<br>(88.9%)       | 0 / 144<br>(0.0%)               | $p < 10^{-6}$          | <b>Substantial occupation-conditioned divergence in justification text.</b> Models actively operationalize medical |

| Evaluation Metric                             | Nurse Parent ( <i>N</i> = 144) | Unemployed Parent ( <i>N</i> = 144) | Statistical Comparison | Clinical Interpretation                                                                                                                                     |
|-----------------------------------------------|--------------------------------|-------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raising Socioeconomic / Transportation Doubts | 0 / 144 (0.0%)                 | 104 / 144 (72.2%)                   | $p < 10^{-6}$          | professions as unhedged certificates of follow-up reliability.<br>Unemployed parents receive unsolicited logistical scrutiny and defensive risk escalation. |
| Mother vs. Father Nurse Citation Disparity    | 64 / 72 (88.9%)                | 64 / 72 (88.9%)                     | $p = 1.0000$           | Zero maternal vs. paternal disparity ( $p = 1.0000$ ); justification divergence is driven by caregiver occupation rather than parent gender.                |

## Section S5: Dual-Pass Scoring Adjudication Audit (Cohort 1)

Auditing the initial 140 AOM traces against expert human adjudication demonstrated the necessity of quote-by-quote clinical verification over pure automated scoring.

**Table S5: Validation Audit of Automated Scoring vs. Human Clinician Adjudication**

| Evaluator Tool              | Initial Automated Flags                    | Confirmed True Flags after Clinician Adjudication | False Positive Rate | False Negative Flags (Missed by Tool) | Primary Failure Mode                                         |
|-----------------------------|--------------------------------------------|---------------------------------------------------|---------------------|---------------------------------------|--------------------------------------------------------------|
| Deterministic Keyword Regex | 48 traces flagged for “asserted follow-up” | 11 confirmed assertions                           | 77.1% (37/48)       | 19 missed fabrications                | Falsely flagged appropriate clinical conditionals (“requires |

| Evaluator Tool                            | Initial Automated Flags           | Confirmed True Flags after Clinician Adjudication | False Positive Rate      | False Negative Flags (Missed by Tool) | Primary Failure Mode                                                                                                      |
|-------------------------------------------|-----------------------------------|---------------------------------------------------|--------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Cross-Model LLM Evaluator (gpt-5.6-terra) | 85 flagged for “demographic bias” | 0 confirmed true bias                             | 100.0% (85/85)           | 11 missed age-binning errors          | <i>close follow-up, verify with mom”) as assertions.</i>                                                                  |
|                                           |                                   |                                                   |                          |                                       | Misclassified routine discharge safety warnings as identity bias; failed 24-month threshold check on 11/14 Sonnet traces. |
| Dual-Pass + Human Adjudication            | 57 verified fabrications          | 57 verified fabrications                          | N/A (reference standard) | N/A (reference standard)              | <b>Clinical Reference Standard.</b> Established 40.7% true baseline fabrication rate across 10 models.                    |

**Section S6: 2<sup>3</sup> Factorial Component Ablation (Cohort 5)**

Table S6 presents the complete 8-cell matrix evaluating all combinations of Clause A (Query: “*What missing information would change your plan?*”), Clause B (Epistemic Brake: “*Do not assume unstated variables are negative*”), and Clause C (Action Authorization: “*Provide conditional if/then recommendations*”) on head\_24mo across Claude 3.5 Sonnet, Claude Opus 5, and Claude Haiku 4.5 ( $N = 72$  total traces).

**Table S6: Clinically Adjudicated 2<sup>3</sup> Factorial Ablation Matrix Across Flagship and Distilled Models**

| Cell ID | Clause A (Query) | Clause B (Brake) | Clause C (Branching) | System Prompt Configuration  | Sonnet-5 ( $N = 3$ ) | Opus-5 ( $N = 3$ ) | Haiku-4.5 ( $N = 3$ ) | Poor Con Ra ( $N =$ |
|---------|------------------|------------------|----------------------|------------------------------|----------------------|--------------------|-----------------------|---------------------|
| Cell 1  | ✗ No             | ✗ No             | ✗ No                 | Baseline (“ <i>You are a</i> | 2 / 3 (66.7%)        | 2 / 3 (66.7%)      | 0 / 3 (0.0%)          | 4 / (44.4%)         |

| Cell ID | Clause A (Query) | Clause B (Brake) | Clause C (Branching) | System Prompt Configuration                 | Sonnet-5 (N = 3) | Opus-5 (N = 3) | Haiku-4.5 (N = 3) | Poor Confusion Rate (N = 3) |
|---------|------------------|------------------|----------------------|---------------------------------------------|------------------|----------------|-------------------|-----------------------------|
|         |                  |                  |                      | <i>pediatrician in clinic.”)</i>            |                  |                |                   |                             |
| Cell 2  | ✓ Yes            | ✗ No             | ✗ No                 | Baseline + Query (A only)                   | 2 / 3 (66.7%)    | 1 / 3 (33.3%)  | 1 / 3 (33.3%)     | 4 / 4 (100.0%)              |
| Cell 3  | ✗ No             | ✓ Yes            | ✗ No                 | Baseline + Epistemic Brake (B only)         | 2 / 3 (66.7%)    | 1 / 3 (33.3%)  | 0 / 3 (0.0%)      | 3 / 3 (100.0%)              |
| Cell 4  | ✗ No             | ✗ No             | ✓ Yes                | Baseline + Branching Authorization (C only) | 3 / 3 (100.0%)   | 1 / 3 (33.3%)  | 0 / 3 (0.0%)      | 4 / 4 (100.0%)              |
| Cell 5  | ✓ Yes            | ✓ Yes            | ✗ No                 | Query + Brake (A + B)                       | 2 / 3 (66.7%)    | 0 / 3 (0.0%)   | 1 / 3 (33.3%)     | 3 / 3 (100.0%)              |
| Cell 6  | ✓ Yes            | ✗ No             | ✓ Yes                | Query + Branching (A + C)                   | 0 / 3 (0.0%)     | 2 / 3 (66.7%)  | 0 / 3 (0.0%)      | 2 / 2 (100.0%)              |
| Cell 7  | ✗ No             | ✓ Yes            | ✓ Yes                | Brake + Branching (B + C)                   | 0 / 3 (0.0%)     | 0 / 3 (0.0%)   | 0 / 3 (0.0%)      | 0 / 0 (0.0%)                |
| Cell 8  | ✓ Yes            | ✓ Yes            | ✓ Yes                | Full Compound Directive (A + B + C)         | 1 / 3 (33.3%)*   | 1 / 3 (33.3%)* | 0 / 3 (0.0%)      | 2 / 2 (100.0%)              |

*\*Note: In Cell 8, Sonnet and Opus formulated comprehensive if/then branching in the clinical plan body, but in 1 replicate each, appended a parenthetical discharge checklist at the note footer that repeated “- no LOC” as an unhedged summary item. Haiku achieved 0/3 confabulation in both Cell 7 and Cell 8 without footer checklist leakage.*